

Regional Transfer, Geometric Feedback, and Local Memory: A Finite Spherical Realization of Nested-Space Inheritance

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Abstract

We construct a finite spherical realization in which regional state differences, energy exchange, geometric feedback, maintained localization, and a measurable local response occur under one action. Six Gaussian fermion columns and the metric evolve together on a periodic radial chart. A conformal gauge gives wave transport for the continuum constraints and an explicit forcing budget for the projected finite dynamics. Four matched runs reach coordinate time 0.3 without a state reset. A localized region remains the leader while normal energy flows across three natural regional surfaces; the associated space, time, and quadrature indicators are below one percent. Curvature is computed from the realized metric and the actual projected time jet, rather than inferred from an auxiliary field. On the active interval 0.2 to 0.3, reduction to the original two-mode observer reproduces the generated occupation and coherence. Memory, initial exterior drive, and actual initial cross correlations each produce a local effect larger than its independently measured comparison discrepancy. Source-imbalance controls change transfer and leadership under the same law. The construction reuses finite normalized inheritance and standard operator elimination. It demonstrates maintained structure with regional transport and local memory on a declared finite realization, with numerical precision indicators; it does not infer a periodic return, new matter, an eternal continuation, or a continuum state-error certificate.

1 The finite result and its organizing relation

The question is whether a regional difference can support exchange, state–geometry feedback, maintained structure, and a local measurement within the same equations. We exhibit this chain on a finite periodic spherical chart. The result combines a coupled forward trajectory with an exact form of reduced dynamics and quantified numerical comparisons. The spatial window remains populated and carries energy through its boundaries while the original mode observer changes substantially.

The organizing relation of Nested-Space Cosmology is

$$\mathcal{T}_\Theta^* \mathbb{D}_\Theta = \mathbb{D}_\Theta. \quad (1)$$

The preserved foundation provides the physical motivation and the one-spectrum formulation [1]. Its finite operator companion specifies normalized inheritance [2]. For a Hermitian prototype H , link B , scale $\Omega > 0$, and finite depth N , the blocks $(J_{m,N})_{jj} = \Omega^{m+j} H$ and $(J_{m,N})_{j,j+1} = \Omega^{m+j} B$ obey

$$J_{m+1,N} = \Omega U_m J_{m,N} U_m^\dagger, \quad \text{spec}(\Omega^{-m} J_{m,N}) = \text{spec}(J_{0,N}). \quad (2)$$

Here U_m relabels corresponding equal-depth windows. Every block gains one factor Ω , which proves the identity; unitary spectral covariance then proves the second statement. Standard ordered Schur elimination preserves the response of the removed blocks.

Table 1: Fixed geometric inputs of the selected local action. Values are model inputs from the retained same-spectrum record, in reference units.

Coefficient	Value	Coefficient	Value
A	0.04501936182826115	C_W	-0.0008441130342798952
C_F	0.01125484045706527	b_{mag}	4
C_E	0.0005158468542821582	$C_{\square}$	-0.0005627420228532635
V_{rel}	0	κ	1

We reuse this finite distinction between inherited law and regional state. The present realization fixes one action and source inventory, while occupations and geometry can differ. Its evolving spatial windows are not asserted to equal a frozen, hand-selected H, B hierarchy. Geometry changes the generated Dirac operator and its dressed spectrum. Equation (2) therefore supplies an operator-level construction, not an additional measured cosmological scaling law.

2 Declared action, chart, and Gaussian source

We use dimensionless reference units with $\hbar = 1$, radial period $\mathcal{L} = 8$, and the spherical metric

$$ds^2 = r^2 [L^2 dt^2 - Q^2(dx + \beta dt)^2 - d\Omega_2^2]. \quad (3)$$

The physical lapse and radial metric coefficient are $N = rL$ and $q = rQ$. The positive chart has $r, Q, L > 0$. Spatial geometry is periodic and spinors are antiperiodic. This is a finite boundary problem, not an asymptotically flat exterior or a fitted cosmological background.

The selected branch Γ_{one} contains the canonical Gaussian source and a same-spectrum local induced geometric term, counted once. Spectral-action and heat-kernel methods provide its established background [3, 4]; spherical canonical reduction is standard [5]. We adopt the recorded local truncation and coefficients, without claiming a UV completion or a newly determined set of fundamental constants. After the owned boundary completion, its first-order geometric density is

$$\begin{aligned} \mathcal{L}_g &= 2\mathcal{K}(F_ru + fv) - 2F_x L_x / Q + Z [(Q/L)u^2 - (L/Q)r_x^2] + LQV, \\ \mathcal{K} &= [Q_t - (\beta Q)_x] / L, \quad u = r_t - \beta r_x, \quad v = \chi_t - \beta \chi_x, \\ F &= -4\pi A r^2 + f\chi, \quad f = -8\pi C_W / 3, \quad Z = -24\pi A, \\ V &= 8\pi A r^2 - \frac{f}{2}(4\chi + \chi^2) - 2\pi C_F b_{\text{mag}}^2. \end{aligned} \quad (4)$$

Time-cap data are Q, r, χ ; the spatial boundary flux integrates to zero. The Euler and total-divergence terms remain boundary terms on this domain. The auxiliary χ linearizes the Weyl term. Its equation supplies $\chi = R_h - 2$ in the continuum action, but the realized metric curvature is calculated independently in Section 5.

The nonsingular first-order chart requires $A \neq 0$ and $C_W \neq 0$. The magnetic convention is $F_{\theta\phi} = b_{\text{mag}} \sin \theta / 2$. The retained input gives $C_F b_{\text{mag}}^2 / (4A) = 1$; V_{rel} is not added again to (4).

The canonical spinor is the half-density $\varphi = r\sqrt{q}\psi$. One representative angular block has

$$H(L, Q, \beta) = \sigma_2 \frac{1}{2} \{L/Q, P\} + \kappa L \sigma_1 - \frac{1}{2} \{\beta, P\}, \quad P = -i\partial_x. \quad (5)$$

Its finite Gaussian covariance is

$$\mathbf{C} = \Phi \text{diag}(c) \Phi^\dagger, \quad c = (0.75, 0.75, 0.5, 0.5, 0.25, 0.25), \quad \Phi^\dagger \Phi = I_6. \quad (6)$$

Thus $0 \leq \mathbf{C} \leq I$ and $\text{Tr } \mathbf{C} = 3$. Zero occupation of the remaining band is a finite preparation, not an identification of a physical vacuum. Gaussian one-body expressions use established fermionic trace identities [6].

The source consists of three real two-lobe envelopes. With $b(u) = \exp[-1/(u(1-u))]$ on $0 < u < 1$ and zero otherwise, the two lobes are normalized $b(u)$ and $(u - 1/2)b(u)$. Packet n combines the even lobe on $[n, n+1]$ with the odd lobe on $[n+1, n+2]$, followed by real symmetric orthonormalization. Its plus column has local carrier $e^{ik(x-n)}f_n(x)(1, i)^T/\sqrt{2}$, with $k = 1$. The minus column is $e^{i\vartheta}$ times its conjugate, $\vartheta = -0.842669616901$. This constant column phase does not change $\mathbf{C}$, but the original observer phases are retained for coherence. The half-density samples include $\sqrt{\Delta x}$.

In this truncation, the state-resolved variational representation is

$$\Gamma_{\text{one}} = \int \mathcal{L}_g dt dx + M \sum_a c_a \int dt \left\langle \varphi_a, \left(\frac{i}{2} \overleftrightarrow{\partial}_t - H \right) \varphi_a \right\rangle_x. \quad (7)$$

The symmetrized time term gives the same canonical field equation and a real functional. Fixed nonzero weights cancel from that equation, while remaining in the metric variation. This is the selected local/state branch, not a second geometric source supplied by the later exterior elimination. The angular copy weight is $M = 4\kappa = 4$. It multiplies geometric source energy and force,

$$E_f = M \sum_a c_a \langle \Phi_a, H_G \Phi_a \rangle, \quad (8)$$

while each representative column evolves by $\dot{\Phi}_a = -iH_G \Phi_a$. There is no additional M in this generator or in a modal occupation.

3 Constraints, conformal formulation, and finite projection

The momenta derived from (4) give $\Pi = p_r - (F_r/f)p_\chi$. Before choosing lapse and shift, their variations retain the constraints

$$\begin{aligned} \mathcal{C}_g &= \frac{p_Q p_\chi}{2f} + \frac{\Pi^2}{4ZQ} + \frac{Zr_x^2}{Q} - QV - 2\partial_x(F_x/Q), \\ \mathcal{D}_g &= p_r r_x + p_\chi \chi_x - Q\partial_x p_Q. \end{aligned} \quad (9)$$

Define the continuum column moments

$$\begin{aligned} K &= \sum_a c_a \text{Re}(\varphi_a^\dagger \sigma_2 P \varphi_a), \\ S_1 &= \sum_a c_a \varphi_a^\dagger \sigma_1 \varphi_a, \\ J &= \sum_a c_a \text{Re}(\varphi_a^\dagger P \varphi_a). \end{aligned}$$

The source energy derivatives per coordinate length are

$$\mathfrak{f}_L = M(K/Q + \kappa S_1), \quad \mathfrak{f}_Q = -MLK/Q^2, \quad \mathfrak{f}_\beta = -MJ. \quad (10)$$

The total constraints are $\mathcal{C} = \mathcal{C}_g + \mathfrak{f}_L$ and $\mathcal{D} = \mathcal{D}_g + \mathfrak{f}_\beta$. Source and geometry therefore belong to the same Hamiltonian variation.

We choose $L = Q$, $\beta = 0$ after retaining those constraints. The Hamiltonian is $H_c = \int h_c dx$, with $h_c = QC$. Differentiation of this Q -dependent lapse matters: relative to fixed-lapse rates evaluated at $L = Q$, the p_Q rate gains $-\mathcal{C}_g$, and its matter contribution is $-(\mathfrak{f}_Q + \mathfrak{f}_L)$. This is the chain rule of the gauge-fixed action; it vanishes as an extra constraint term on the constrained surface. It is not a state correction or a new radius force. The field generator simplifies to

$$H_c^{(f)} = \sigma_2 P + \kappa Q \sigma_1. \quad (11)$$

The kinetic coefficient is one; geometry remains in the angular mass term and in the physical metric.

Proposition 1 (Continuum constraint transport). *For smooth periodic fields in the positive chart, the Hamiltonian $\int QC dx$ derived above satisfies*

$$\partial_t h_c = \partial_x \mathcal{D}, \quad \partial_t \mathcal{D} = \partial_x h_c. \quad (12)$$

Proof. Set $a = F_{rr} = -8\pi A$, so $F_r = ar$ and $Z = 3a$. Differentiate (9) with the canonical equations in Appendix A. The spatial derivatives in F_x/Q supply the improved momentum density $-Qp_{Q,x}$. The field equations $i\varphi_t = (\sigma_2 P + \kappa Q \sigma_1)\varphi$ give the derivatives of K, S_1, J in (10). Their angular-mass terms cancel the $-M\kappa S_1$ term in $\dot{p}_Q$. The remaining terms are respectively $\partial_x \mathcal{D}$ and $\partial_x (QC)$. This calculation uses $V_r = -2ar$ and $V_\chi = -f(2 + \chi)$; the constant magnetic term differentiates to zero. It applies column by column and therefore to arbitrary fixed Gaussian weights. Independent symbolic Euler derivatives and directional checks reproduce the identities. $\square$

Consequently vanishing smooth constraints propagate homogeneously. The numerical projection is kept distinct from this continuum identity.

Geometry uses the odd periodic band $n_g = n_f - 1$ and the columns use the even antiperiodic band n_f . Both are prolonged to $n_q = 4n_f$ quadrature nodes. With A_g, A_f the nodal interpolation maps,

$$P_g = \frac{n_g}{n_q} A_g^T, \quad U_f = \sqrt{n_f/n_q} A_f, \quad P_g A_g = I, \quad U_f^\dagger U_f = I. \quad (13)$$

The fine Hamiltonian is evaluated on the prolonged state, and every geometric and field rate is pulled back by the exact respective adjoint. In particular $H_G = U_f^\dagger H_c^{(f)} U_f$ is Hermitian. Products use the summation-by-parts energy and the actual discrete $D_x F$, not a silently imposed continuum product rule.

For the actual finite ODE, define the full quadrature defects

$$f_h = \dot{h}_c - D_x \mathcal{D}, \quad f_D = \dot{\mathcal{D}} - D_x h_c, \quad R^2 = \Delta x_q \sum (h_c^2 + \mathcal{D}^2). \quad (14)$$

Skew adjointness of D_x yields

$$\frac{dR^2}{dt} = 2\Delta x_q \sum (h_c f_h + \mathcal{D} f_D), \quad R(t) \leq R(0) + \int_0^t \|(f_h, f_D)\|_2 ds. \quad (15)$$

Analytic directional derivatives of the actual sampled constraints expose both unprojected quadrature forcing and the additional rate-projection forcing. Neither is discarded. Equation (15) is a finite ODE estimate without an exponential in the Fourier cutoff. Its sampled time integral is a numerical indicator, not a certified continuum enclosure.

Table 2: Initial and final full constraint diagnostics. These are sampled norms, separately reported from projected solver residuals.

n_f , step cap	$\ \mathcal{C}(0)\ _\infty$	$R(0)$	$R(0.3)$	$R_{\text{budget}}(0.3)$
256, 0.0005	1.7681×10^{-3}	2.9599×10^{-4}	2.9761×10^{-4}	3.9497×10^{-4}
512, 0.0005	9.9815×10^{-6}	2.0955×10^{-6}	2.0862×10^{-6}	4.3987×10^{-4}

4 Forward episodes and maintained regional transfer

The initial $Q = 0.251324933269135$ is constant; χ and the momenta start at zero in the retained baseline. The radius is the saved source-derived solution of the projected lapse constraint. Initial source and intrinsic and extrinsic data do not change when choosing the conformal gauge, because f_L, f_β are independent of L, β . The near-zero initial current is reported rather than subtracted.

Four runs use $n_f = 256, 512$ and step caps 0.001, 0.0005. The first episode reaches $T = 0.05$; the second continues each corresponding Cauchy state bitwise to $T = 0.3$. Each RK4 stage recomputes the source, geometric rates, and Dirac image on that stage state. The next step is bounded by $1.4/\omega$, where $\omega = k_{\max} + \kappa \max Q$, and by the output-frame endpoint. All four runs remain in the positive chart with column Gram errors below 5.3×10^{-12} ; the fine half-step case stays below 1.7×10^{-13} during the continuation.

Table 2 states the initial residual domain explicitly. The fine endpoint has full maxima $\|\mathcal{C}\|_\infty \simeq 9.00 \times 10^{-6}$ and $\|\mathcal{D}\|_\infty \simeq 7.98 \times 10^{-7}$. The forcing integral is not asserted to decrease with resolution: its fine joined value is about 4.38×10^{-4} , dominated by rate projection. This constraint-control primitive and the physical refinement indicators answer different questions; the initial full residual is part of the declared numerical domain.

The four natural spatial windows are $I_j = [2j, 2j+2]$, $j = 0, \dots, 3$. Contents are exact integrals of the periodic nodal trigonometric interpolants on those intervals. Let $V_x = 4\pi r^2 q$ be physical volume per coordinate length. The normal-observer shell and canonical probability are

$$E_N(I) = \int_I V_x \rho_N dx, \quad \mathcal{P}(I) = \sum_a c_a \int_I \varphi_a^\dagger \varphi_a dx. \quad (16)$$

The normal shell is the nodal sum of F_L/r . This differs from the coordinate Hamiltonian energy and from a two-mode occupation.

Region 0 remains the normal-shell and probability leader throughout. During the continuation its shares stay above 0.65098 and 0.65585. Its normal shell changes from 6.62107371 at 0.05 to 6.57976032 at 0.3; its probability content falls by 0.03246364. Maintenance here means persistence of the localized leader during regional transport. Its shell and probability contents decrease; the individual leader-surface flows can both point outward. Equal entry and exit, constant content, circulation, and recovery of an identical state are not inferred. The finite interval was extended because outer-boundary flow at 0.05 was unresolved, despite resolved internal current.

For outward normal-energy flux $\mathcal{J}_N$, each surface is assessed individually through $\int |\mathcal{J}_N| dt$ as well as its signed integral. Opposing nonzero entry and exit can cancel in a net-content balance and are not rejected for that reason. The fourth distinct surface, $x = 6$, is unresolved on this realization. The three reported exchanges satisfy the declared one-percent numerical criterion in their stated windows; that label is not applied to every constraint diagnostic or to a total continuum error.

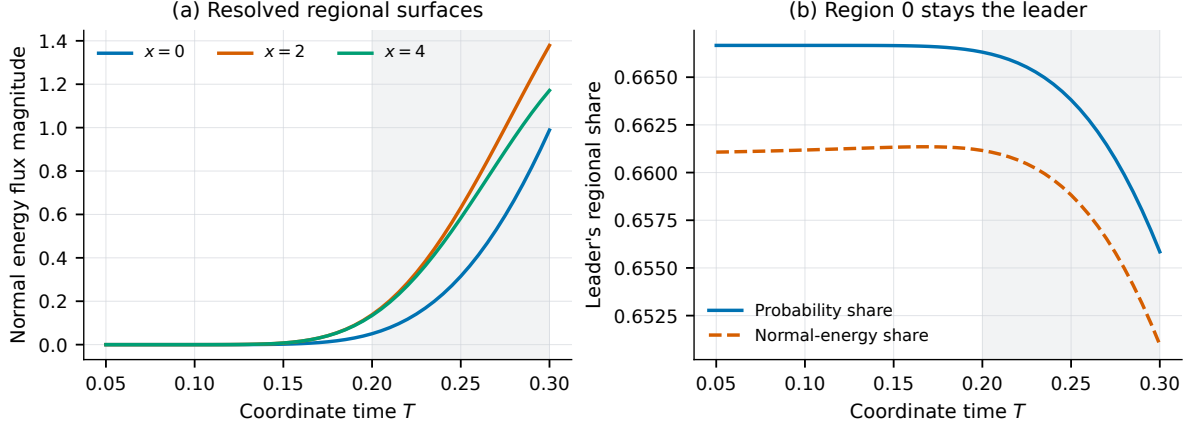


Figure 1: Continuation of the same realization on $0.05 \leq T \leq 0.3$: individual physical surface flows and the region-0 shell and probability shares. The handoff at 0.05 continues the full Cauchy state of the first episode. Energy flow is separated from canonical probability and from the fixed mode-observer measurement.

Table 3: Absolute normal-energy-flow integrals during $T \in [0.05, 0.3]$. Indicators are movement divided by each surface's own flow effect.

Surface	Integral	time indicator	space indicator	frame quadrature
$x = 0$	0.03937795	$4.41 \times 10^{-7}\%$	0.00214%	0.09503%
$x = 2$	0.07025487	$5.81 \times 10^{-7}\%$	0.00636%	0.04273%
$x = 4$	0.06351828	$6.17 \times 10^{-7}\%$	0.00134%	0.03139%

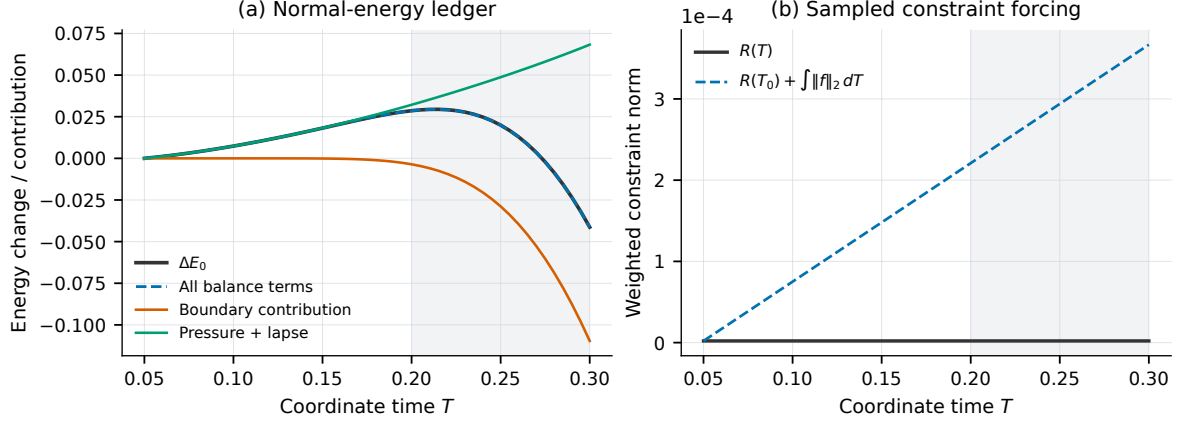


Figure 2: Normal-observer shell accounting (left): boundary flow, pressure work and lapse-gradient work are distinct from coordinate field–geometry energy exchange. The sampled constraint norm and integrated forcing budget (right) use (15); the sampled integral is not a certified continuum or temporal bound.

4.1 Coordinate work and the normal shell ledger

The geometry exchanges energy with the field through the same variation. If F_i are nodal energy derivatives, coordinate work is

$$\dot{E}_f = \sum (F_Q \dot{Q} + F_L \dot{L} + F_\beta \dot{\beta}) = \sum (F_Q + F_L) \dot{Q}_{\text{lifted}}. \quad (17)$$

There is no extra spacing or angular multiplicity in this nodal sum. Using only $F_Q \dot{Q}$ would omit the evolving lapse. Across both fine episodes the field-energy change is about -2.37443×10^{-5} , balanced by the geometric change to 1.31×10^{-11} .

For the normal observer, define expansion rates $K_r = [q_t - (\beta q)_x]/(Nq)$ and $K_\perp = (r_t - \beta r_x)/(Nr)$. The continuum shell balance is

$$\begin{aligned} \partial_t(V_x \rho_N) + \partial_x \mathcal{J}_N &= \mathcal{W}_p + \mathcal{W}_N, \\ \mathcal{J}_N &= V_x(Nj_N/q - \beta \rho_N), \\ \mathcal{W}_p &= -NV_x(p_r^{(N)} K_r + 2p_\perp^{(N)} K_\perp), \\ \mathcal{W}_N &= -V_x(j_N/q)N_x. \end{aligned} \quad (18)$$

The source derivatives give the stresses, for example $\rho_N = f_L/(4\pi r^4 Q)$ and $j_N = -f_\beta/(4\pi r^4 Q^2)$. The realized finite balance includes its measured directional residual, which is retained rather than zeroed. For region 0 on $[0.05, 0.3]$, frame-Simpson integrals of net boundary flow, pressure work, and lapse-gradient work are approximately -0.10956538 , $+0.04884147$, and $+0.01941054$. They account for the shell change -0.04131339 to about 1.99×10^{-8} ; the trapezoid balance indicator is about 0.164% of that shell change.

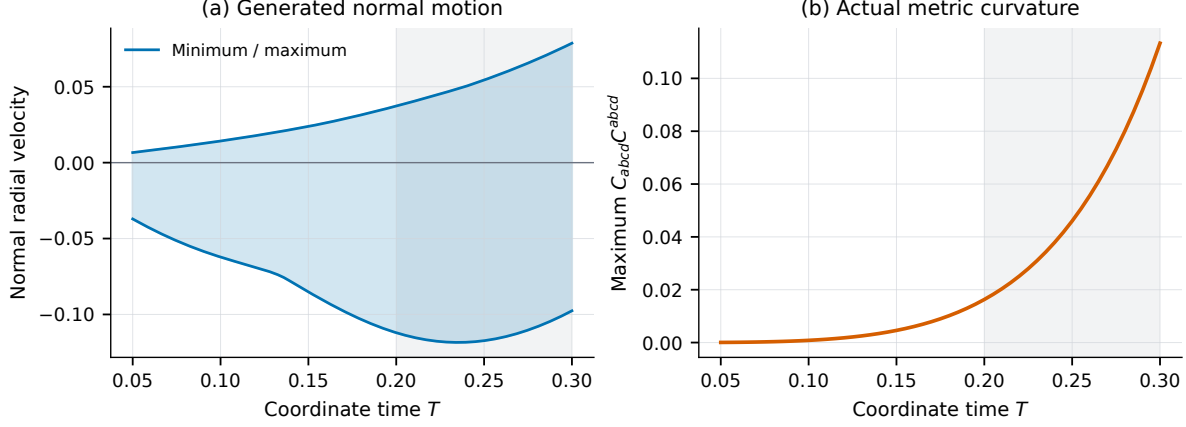


Figure 3: Normal radial motion and actual metric curvature on the generated positive chart. The Weyl invariant is computed from the projected ODE time jet and spatial metric jet; agreement with the auxiliary proxy is assessed afterward. The endpoint proper-clock control uses this same metric.

5 Realized curvature and the proper-clock control

The metric itself supplies an invariant measurement. In the conformal chart, $\dot{Q}_g = P_g[Q_f p_{\chi,f}/(2f)]$. Differentiating this actual finite rate along the actual projected ODE gives

$$\ddot{Q}_f = A_g P_g \left[\frac{\dot{Q}_f p_{\chi,f} + Q_f \dot{p}_{\chi,f}}{2f} \right], \quad \dot{Q}_f = A_g \dot{Q}_g, \quad \dot{p}_{\chi,f} = A_g \dot{p}_{\chi,g}. \quad (19)$$

Both projections and $\dot{L} = \dot{Q}_f$ are kept. The two-metric scalar and spherical Weyl invariant are

$$R_h = \frac{2}{Q^2} \left[\frac{Q_{xx}}{Q} - \frac{Q_x^2}{Q^2} - \frac{\ddot{Q}}{Q} + \frac{\dot{Q}^2}{Q^2} \right], \quad \mathcal{W} = C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} = \frac{(R_h - 2)^2}{3r^4}. \quad (20)$$

The second identity is the owned conformal spherical product relation. Directional differences of the finite rate validate (19); an independent Christoffel contraction validates the metric sign and factors.

At the fine $T = 0.3$ endpoint, $R_h \in [11.45342664, 15.71659763]$ and $\max \mathcal{W} = 0.11331608321$. Its spatial indicator is 0.00110%, and its timestep movement is about 5.13×10^{-11} . Only after this calculation is the auxiliary relation compared: $\max |R_h - (\chi + 2)| \simeq 7.88 \times 10^{-8}$ and the maximum pointwise Weyl/proxy gap is about 1.29×10^{-9} . The dense and FFT realizations of the same spatial derivative differ in finite precision, with a recorded endpoint curvature-formula gap 2.64×10^{-6} . This conditioning indicator is distinguished from an exact-arithmetic identity or a continuum enclosure.

The reference normal worldline is fixed at $x = 1$, with $d\tau = rQdt$. Its clock is continuous from $T = 0$ and reaches $\tau = 0.3616114057$ at $T = 0.3$. The frozen-geometry control keeps the original Q_0 and columns. Its constant generator has modal frequencies $\sqrt{p^2 + \kappa^2 Q_0^2}$, so its propagation is evaluated analytically. At the same normal-clock endpoint, the original observer's two occupation separations are approximately $(-1.19109, -1.06846) \times 10^{-5}$. Their time and space indicators are below 0.0008% and 0.00014%. No interpolation of the frozen control or reset of the clock is required. Most occupation change is also present in frozen geometry; this smaller resolved separation identifies the measured geometric contribution under the declared observer and clock protocol.

6 Source-imbalance controls under the same law

The source family

$$c(\alpha) = (\frac{1}{2} + \frac{1}{4}\alpha, \frac{1}{2} + \frac{1}{4}\alpha, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} - \frac{1}{4}\alpha, \frac{1}{2} - \frac{1}{4}\alpha) \quad (21)$$

preserves total occupation three and equal weights inside each conjugate pair. Four exploratory coarse runs use $\alpha = 0, -1, 0.95, 1.05$. The original columns are unchanged; the source radius and periodic shift momentum are solved anew at the initial slice, and the current mean remains in the shift residual. Full initial Hamilton maxima range from 0.00137 to 0.00179 and retain their declared solver-floor domain. All four runs reach 0.3, keep a positive chart and Gram gaps below 1.7×10^{-13} , and have positive sampled forcing-budget margins.

Reversal changes the signed transfer at $x = 2$ from about +0.07025 to -0.07024 and moves the maintained normal-shell leader to region 1. The $\pm 5\%$ imbalance perturbations keep region 0 as leader and change that transfer by approximately $\pm 5.04\%$. The changes at $x = 4$ are approximately $\mp 4.86\%$. Paired frame-quadrature indicators are about 0.043% and 0.031% of those measured differences. These coarse controls establish source sensitivity; the baseline's fine comparisons are reference scales, not a control-specific continuum error certificate.

Uniform weights leave substantial outer flow: about 0.02664 at $x = 0$ and 0.12415 at $x = 4$. They represent half of a rank-six projector, not half the identity on the full band. The empty complement remains available to the dynamics. The uniform run has two comparably occupied windows and changes its normal-shell leader; no minimum-share or reversal condition is imposed to hide that outcome.

7 The same realization seen by a fixed mode observer

A spatial energy window and a Hilbert-space observer are distinct. Let V be the original $T = 0$ columns 0 and 1, preserved without QR, phase change or transport to a new observer. Its projector is $P_A = VV^\dagger$. The complement $P_E = I - P_A$ includes modes with support inside and outside $[0, 2]$. Throughout, E means exterior to the retained *mode space*. The omission controls do not isolate a contribution solely from a geometric parent region or attribute the response to only the $x = 2$ surface. The local measurements are the two diagonal occupations and the complex off-diagonal coherence of $V^\dagger CV$. They are not identified with integrated shell energy or with the window's canonical probability.

The reduction starts with the actual transported $\Phi(0.2)$, not the original columns placed again at the segment boundary. At that point the local covariance eigenvalues are approximately 0.103233, 0.107916, and the actual initial cross norm is $\|C_{AE}\|_F = 0.3386366$. The original weights remain those in (6). Generated Q is prolonged to the original quadrature and linearly interpolated between the 21 stored frames on $[0.2, 0.3]$. This defines the conditional field schedule used by both full and reduced propagation. Independent Fourier actions match the generated Galerkin operator to below 4.4×10^{-13} relative error.

7.1 Exact elimination and actual initial correlations

In the fixed orthonormal frame $[V, V_E]$, write $H_G(t) = \begin{pmatrix} A & B \\ B^\dagger & D_E \end{pmatrix}$. The free exterior propagator obeys $i\partial_t U_E(t, s) = D_E(t)U_E(t, s)$ and $U_E(s, s) = I$. For retained and exterior column amplitudes

Table 4: Causal omissions on the same generated geometry, with each control’s own independent reference error. These are conditional controls, not separate coupled geometric evolutions.

Omission	Occupation movement	Own reference error	Error/effect
Memory	0.0168185383	2.2011×10^{-6}	0.01309%
Initial exterior drive	0.0485412050	1.6626×10^{-5}	0.03425%
Actual initial cross	0.0920614328	1.2282×10^{-5}	0.01334%

x, y , integration of the exterior equation gives

$$\begin{aligned}
y(t) &= U_E(t, t_*)y_* - i \int_{t_*}^t U_E(t, s)B^\dagger(s)x(s)ds, \\
\dot{x}(t) &= -iA(t)x(t) - iB(t)U_E(t, t_*)y_* - \int_{t_*}^t K(t, s)x(s)ds, \\
K(t, s) &= B(t)U_E(t, s)B^\dagger(s), \quad t_* = 0.2.
\end{aligned} \tag{22}$$

Thus unresolved degrees remain as causal memory and initial exterior drive. The initial covariance contributes cross terms as well as its two marginals. For the exact responses R_A, R_E from each initial component,

$$\begin{aligned}
C_A(t) &= R_A C_{AA} R_A^\dagger + R_E C_{EE} R_E^\dagger \\
&\quad + R_A C_{AE} R_E^\dagger + R_E C_{EA} R_A^\dagger.
\end{aligned} \tag{23}$$

This follows directly by expanding the covariance of the summed amplitude. It is standard block elimination, applied here to the generated metric and actual correlated state, rather than a new quantum law.

The streamed solver transports twelve columns: the retained and exterior pieces of each of the six actual columns. Linearity permits their coherent sum to reconstruct the original state. Summing the two component covariances instead removes only the actual initial cross. Retaining only the first component supplies the initial-drive omission. An unsplit pilot and independent full split verify these identities. Time-indexed dense exterior propagators are unnecessary; the stored column history has shape (21, 1022, 12).

7.2 Measured response and individually verified controls

The two autonomous occupations have the endpoints

$$\begin{aligned}
\mathbf{n}(0.2) &= (0.1055744772, 0.1055746375), \\
\mathbf{n}(0.3) &= (0.0321449576, 0.0322689986).
\end{aligned}$$

The maximum occupation and coherence changes are 0.0734295196 and 0.00237420793, respectively.

The streamed/full occupation discrepancy is 6.3508×10^{-6} , or 0.00865% of its effect; the coherence discrepancy is 8.4787×10^{-7} , or 0.03571%. The conditional full occupation differs from the saved autonomous frames by only 2.2951×10^{-10} .

Memory omission is independently compared with the triangular generator $G = H_G - P_E H_G P_A$, which prevents retained-to-exterior feedback while preserving the retained rows. It is explicitly a non-Hermitian negative control, not a replacement physical model. Its two-to-four midpoint-substep indicator is 3.15×10^{-11} . The drive omission is compared with full propagation from the

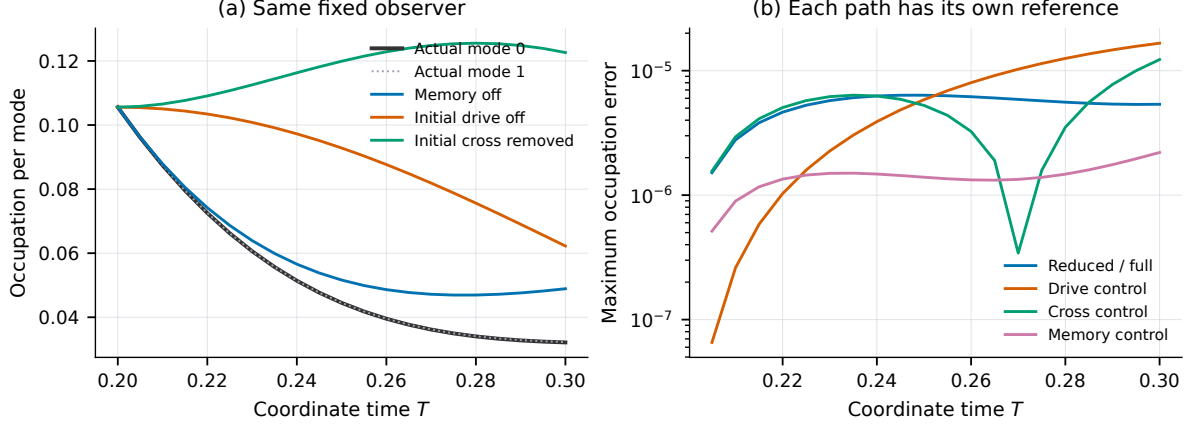


Figure 4: The original fixed observer’s autonomous occupations (modes 0 and 1) and mode-0 omission curves on $T \in [0.2, 0.3]$ (left), using the actual $\Phi(0.2)$ correlations. The modal complement is distinct from the spatial windows. Reduced/full and independent-control comparison errors are maxima over the two modes (right), using Table 4. The initial roundoff sample is omitted from the error panel; coherence is independently quantified in the text.

actual retained initial component. The cross omission is compared with the full propagation of both actual components, with their cross removed only in covariance assembly. No synthetic initial cross is inserted.

Doubling full midpoint substeps and replacing linear Q interpolation with rate-Hermite interpolation using the actual stored $\dot{Q} = \dot{L}$ move occupation and coherence by less than one percent of their respective effects. Geometry-only comparisons retain the primary state and observer while substituting the time- or space-refined Q path on a common quadrature. Separately, the actual autonomous occupation differs by 1.33×10^{-8} across spatial refinement and 2.32×10^{-11} across timestep refinement. These declared-path comparisons do not enclose an unknown unstored metric.

On the same active segment, the three surface values are ordered as $x = 0, 2, 4$. Their absolute normal-energy-flow integrals, largest numerical indicators, and the continuous normal-clock endpoints are

$$\begin{aligned} \text{surface integrals} &= (0.03848436, 0.06757848, 0.06088737), \\ \text{largest indicators} &= (0.08449\%, 0.02699\%, 0.01429\%), \\ (\tau(0.2), \tau(0.3)) &= (0.2412093356, 0.3616114057). \end{aligned}$$

The spatial leader persists while the fixed observer loses occupation, and the unresolved complement actively changes that response. This is the quantitative outside-to-local meaning supported by the calculated memory and initial-state terms.

8 Interpretation, limits, and reproducibility

The established chain is finite: a declared regional Gaussian difference produces transfer; the same action evolves both state and geometry; a localized leader persists with resolved boundary flow; and its generated metric and correlated state support an accurately reproduced local response. The source controls demonstrate sensitivity under the same law, while the frozen-clock comparison isolates a smaller geometric contribution. No periodically renewed state, new matter inventory, or

Table 5: Machine-readable scientific evidence. Basenames are followed by the listed version and a JSON or NPZ extension as applicable.

Component	Record basename	Versions
Episodes	nsc-spherical-conformal-episode	v1, v2
Physical frames	nsc-spherical-conformal-frames	v1
Surface transfer	nsc-spherical-conformal-transport	v2
Proper clock	nsc-spherical-conformal-clock	v2
Source controls	nsc-spherical-conformal-source-controls	v1
Metric invariant	nsc-spherical-conformal-curvature	v1
Local response	nsc-conformal-local-response	v2
Memory reference	nsc-conformal-memory-control	v1

eternal continuation is inferred. “Regeneration” here denotes maintained structure with regional transport over the measured interval.

Analytic and numerical statements are separate. Normalized finite inheritance, Hamiltonian variation, block elimination, and the finite ODE energy identity are algebraic results in stated domains. The trajectory, surface-flow and local-response values are observations with spatial, temporal and quadrature indicators. The latter satisfy the declared one-percent criterion for the reported effects, but are not a total state error certificate against an unknown geometry or a continuum solution. In particular, sampled constraint forcing does not by itself provide the missing physical-observable stability map. A conditional Duhamel estimate illustrates the additional quantity needed: if e is a weighted column-state error under a controlled Hamiltonian mismatch, then

$$e(t) \leq e(t_*) + \int_{t_*}^t \left\| \delta H(s) \Phi(s) c^{1/2} \right\|_F ds, \quad \|\delta C_A(t)\|_F \leq 2a(t)e(t) + e(t)^2, \quad (24)$$

where a is the reference weighted-column norm. The unknown geometric mismatch is not bounded by the present measurements. The local memory result also does not identify its Hilbert-space complement with a new dark stress, antimatter sector, or cosmological abundance.

The machine-readable evidence consists of the versioned JSON/NPZ records in Table 5, under **lab/results/development**. Input, code and payload hashes are bound in the records and the accompanying publication manifest. It pins the scientific snapshot

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The preserved source history distinguishes measured producers from later write-protection wrappers. No sealed trajectory is silently regenerated for the article.

The first episode and its physical postprocessing used about 127 CPU seconds; continuation used 114 and 493 seconds in its coarse and fine pools. The four exploratory source controls used 388 seconds, metric curvature 13 seconds, and local production with its independent controls 84 seconds. The independent memory reference used another 3.64 seconds. These bounded computations forecast cost before expansion and preserve full Cauchy handoffs or compact source-control checkpoints. For read-only local verification, run

```
python scripts/lab.py \
scripts/derive_nsc_conformal_local_response_successor.py --check
```

Scientific tests separately replay operator actions, directional derivatives, source bindings, control identities and saved numerical indicators. The build manifest binds this source, bibliography, fig-

ures, evidence and PDF. The broad foundational manuscript and earlier focused source remain historical documents of the same programme.

Author responsibility and assistance

Substantial AI assistance was used in derivations, implementation, numerical verification, independent checks, and manuscript preparation. Douglas Ek is the sole author and is responsible for the scientific claims, evidence selection, and final manuscript.

A Continuum equations of the conformal Hamiltonian

For $a = -8\pi A$, $F_r = ar$, $Z = 3a$, and $\Pi = p_r - arp_\chi/f$, the geometric velocities are

$$\dot{Q} = Qp_\chi/(2f), \quad \dot{r} = \Pi/(2Z), \quad \dot{\chi} = Qp_Q/(2f) - (ar/f)\Pi/(2Z). \quad (25)$$

The momentum equations, including the same matter source, are

$$\dot{p}_Q = -p_Qp_\chi/(2f) + 2QV + 2F_{xx}/Q - M\kappa S_1, \quad (26)$$

$$\dot{p}_r = \Pi ap_\chi/(2Zf) + 2Zr_{xx} + Q^2V_r + ar\partial_x(2Q_x/Q), \quad (27)$$

$$\dot{p}_\chi = Q^2V_\chi + f\partial_x(2Q_x/Q), \quad (28)$$

with $i\varphi_t = (\sigma_2 P + \kappa Q \sigma_1)\varphi$. For a single column $\varphi = (u + iv, w + iz)^T$, this field equation reads $\dot{u} = -z_x + \kappa Qz$, $\dot{v} = w_x - \kappa Qw$, $\dot{w} = v_x + \kappa Qv$, $\dot{z} = -u_x - \kappa Qu$. It gives the bilinears $K = -uw_x - vz_x + wu_x + zv_x$, $S_1 = 2(uw + vz)$ and $J = uv_x - vu_x + wz_x - zw_x$, weighted and summed over columns. Substitution gives Proposition 1. The displayed $2F_{xx}/Q$ is a continuum product-rule simplification; the finite implementation differentiates its summation-by-parts energy and retains $D_x F$ and the sampled constraint term before pullback.

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